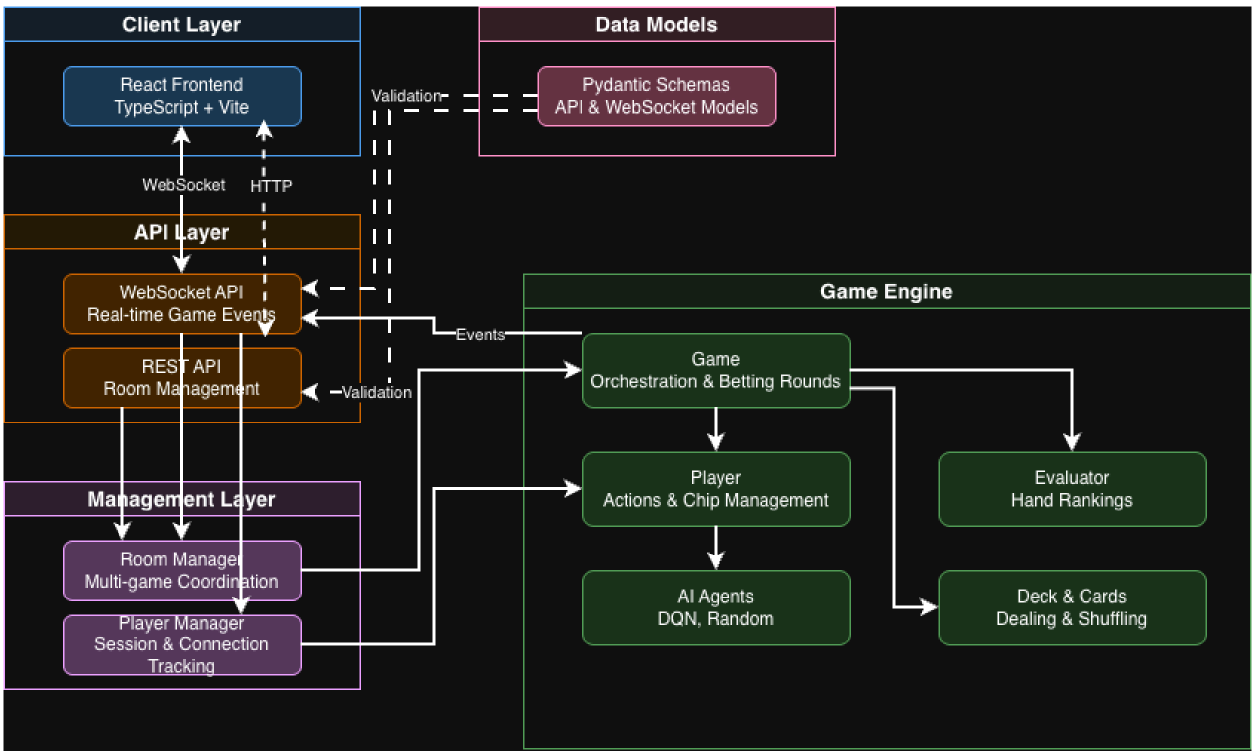
5.0 Requirements Specification

5.1 Introduction

This section of the document specifies the functional and performance requirements for Poker Face 2. This project is a web-based Texas Hold'em Poker application that pits human players against advanced AI agents in real-time. The system migrates from a desktop environment to a web-based architecture, utilizing a FastAPI backend for state management and AI inference, and a React and Vite frontend for the user interface.



The remainder of this document is structured as follows. Section 5.2 contains the Functional Requirements, detailing the breakdown of Computer Software Components (CSCs) for lobby management, game logic, user interface, and AI behavior. Section 5.3 contains the Performance Requirements, outlining the necessary response times, stability, and concurrency standards for the application. Section 5.4 contains the Environment Requirements, specifying the hardware and software stack required for the development and execution of the system.

5.2 Functional Requirements

5.2.1 Lobby & Room Management

F-1.1 As a player, I want to retrieve a list of active game rooms via a REST API call so that I can see which tables are currently available to join.

F-1.2 As a player, I want to create a new game room with specific configurations so that I can start a private or new public game if existing ones are full.

F-1.3 As a player, I want to join a specific room via WebSocket using a unique Room ID so that I can establish a persistent, real-time connection to a game instance.

F-1.4 As a System Admin, I want to delete empty or stale rooms via the API so that I can free up server resources when a game session has concluded.

5.2.2 Game Logic & Rules

F-2.1 As a player, I want to be dealt two private hole cards at the start of a hand so that I can begin assessing my strategy for the round.

F-2.2 As a player, I want to have the system enforce standard Texas Hold'em rules so that I cannot make illegal moves like checking when a bet has already been placed.

F-2.3 As a player, I want to see community cards dealt in the correct stages (Flop, Turn, River) so that the game progresses naturally according to standard poker phases.

F-2.4 As a player, I want to have the pot awarded automatically at showdown or when all others fold so that the winnings are distributed accurately without manual calculation.

F-2.5 As a player, I want to have my chip count updated immediately after betting or winning so thatI always know my exact standing in the game.

5.2.3 User Interface

F-3.1 As a player, I want to see a graphical representation of the table so that I can easily visualize player positions, chips, and the current state of play.

F-3.2 As a player, I want to use action buttons (Fold, Call, Check, Raise) so that I can input my decisions quickly without using text commands.

F-3.3 As a player, I want to use a slider or input box for raising so that I can bet a specific custom amount rather than fixed increments.

F-3.4 As a player, I want to receive visual cues (e.g., highlighting) when it is my turn so that I don't accidentally delay the game by not realizing it is my turn.

F-3.5 As a player, I want to see the current pot size clearly displayed so that I can calculate my pot odds before making a calling decision.

5.2.4 AI Agent Behavior

F-4.1 As a competitive player, I want to play against a Reinforcement Learning Agent so that I can test my skills against an AI that adapts its play style to optimize decisions.

F-4.2 As a competitive player, I want to play against a variety of agent types so that I can have a baseline opponent or a chaotic game experience.

F-4.3 As a competitive player, I want to see the AI bluff occasionally so that the game feels realistic, as the AI isn't just playing based on pure hand strength.

F-4.4 As a developer, I want to load AI models at server startup so that the agents are ready to play immediately without loading delays during the game.

5.3 Performance Requirements

P-1 As a player, I want the UI to update within 100 milliseconds of my action so that the game feels responsive and I don't double-click buttons by mistake.

P-2 As a player, I want the AI to make decisions within 2 seconds so that the flow of the game mimics a human's thinking time without becoming tedious.

P-3 As a developer, I want the server to support multiple concurrent rooms so that several groups of players can play simultaneously without server slowdowns.

P-4 As a developer, I want the system to remain stable for 100+ consecutive hands so that players can enjoy long sessions without crashes or memory leaks.

P-5 As a developer, I want WebSocket messages to broadcast efficiently so that all players see the "Card Dealt" or "Bet Placed" events at virtually the same time.

5.4 Environment Requirements

E-1 As a developer, I want the system to run on Python 3.11+ so that we can utilize the latest features for the FastAPI backend and async handling.

E-2 As a developer, I want the system to utilize PyTorch 1.9+ so that we can support the advanced tensor operations required for the DQN agent.

E-3 As a developer, I want the system to run on a machine with a dedicated GPU so that the AI training and inference steps are accelerated for smoother gameplay.

E-4 As a developer, I want the system to be built using React and Vite so that the client is optimized for modern web browsers and fast development builds.

E-5 As a developer, I want the system to use standard WebSockets for game communication so that the game is compatible with any modern browser (Chrome, Firefox, Safari) without plugins.